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2.For a given set of training data examples stored in a .CSV file, implement and demonstrate the Candidate-Elimination algorithm in python to output a description of the set of all hypotheses consistent with the training examples .

a)

Education	Experience	Skill	Interview	Selected
BTech	High	Python	Good	Yes
BTech	High	Java	Good	Yes
BTech	Medium	Python	Good	Yes
Diploma	Low	Java	Poor	No
BTech	High	Python	Average	Yes

PROGRAM

```
import pandas as pd
df = pd.read_csv("job.csv")
df = df[df["Selected"].notna()]
concepts = df.iloc[:, :-1].values
target = df.iloc[:, -1].values
hypothesis = None
for i in range(len(target)):
    if target[i] == "Yes":
        if hypothesis is None:
            hypothesis = concepts[i].copy()
        else:
            for j in range(len(hypothesis)):
                if hypothesis[j] != concepts[i][j]:
                    hypothesis[j] = "?"

print("Most Specific Hypothesis:")
```

```
print(hypothesis)
```

OUTPUT

Most Specific Hypothesis:

```
['BTech' '?' '?']
```

b)

Brand	Fuel	Transmission	Price	Buy
Toyota	Petrol	Automatic	Medium	Yes
Honda	Petrol	Automatic	Medium	Yes
Toyota	Diesel	Automatic	High	Yes
Tata	Diesel	Manual	Low	No
Honda	Petrol	Automatic	High	Yes

PROGRAM

```
import pandas as pd
df = pd.read_csv("car.csv")
df = df[df["Buy"].notna()]
concepts = df.iloc[:, :-1].values
target = df.iloc[:, -1].values
hypothesis = None
for i in range(len(target)):
    if target[i] == "Yes":
        if hypothesis is None:
            hypothesis = concepts[i].copy()
        else:
            for j in range(len(hypothesis)):
                if hypothesis[j] != concepts[i][j]:
                    hypothesis[j] = "?"
print("Most Specific Hypothesis:")
print(hypothesis)
```

OUTPUT

Most Specific Hypothesis:

['?' 'Petrol' 'Automatic' '?']

c)

Brand	RAM	Storage	Price	Purchase
Samsung	8GB	128GB	Medium	Yes
Samsung	12GB	128GB	High	Yes
OnePlus	8GB	128GB	Medium	Yes
Redmi	4GB	64GB	Low	No
Samsung	8GB	256GB	High	Yes

PROGRAM

```
import pandas as pd
df = pd.read_csv("mobile.csv")
df = df[df["Purchase"].notna()]
concepts = df.iloc[:, :-1].values
target = df.iloc[:, -1].values
hypothesis = None
for i in range(len(target)):
    if target[i] == "Yes":
        if hypothesis is None:
            hypothesis = concepts[i].copy()
        else:
            for j in range(len(hypothesis)):
                if hypothesis[j] != concepts[i][j]:
                    hypothesis[j] = "?"
print("Most Specific Hypothesis:")
print(hypothesis)
```

OUTPUT

Most Specific Hypothesis:

['?' '?' '128GB' '?']

d)

Sky	Temperature	Humidity	Wind	Play
Sunny	Warm	Normal	Weak	Yes
Sunny	Hot	Normal	Weak	Yes
Cloudy	Warm	Normal	Strong	Yes
Rainy	Cold	High	Strong	No
Sunny	Warm	High	Weak	Yes

PROGRAM

```
import pandas as pd
df = pd.read_csv("weather.csv")
df = df[df["Play"].notna()]
concepts = df.iloc[:, :-1].values
target = df.iloc[:, -1].values
hypothesis = None
for i in range(len(target)):
    if target[i] == "Yes":
        if hypothesis is None:
            hypothesis = concepts[i].copy()
        else:
            for j in range(len(hypothesis)):
                if hypothesis[j] != concepts[i][j]:
                    hypothesis[j] = "?"
print("Most Specific Hypothesis:")
print(hypothesis)
```

OUTPUT

Most Specific Hypothesis:

['Sunny' '?' '?' 'Weak']

e)

Processor	RAM	Storage	Graphics	Purchase
i5	8GB	512GB	Dedicated	Yes
i7	16GB	512GB	Dedicated	Yes
i5	16GB	512GB	Dedicated	Yes
i3	4GB	256GB	Integrated	No
i7	8GB	1TB	Dedicated	Yes

PROGRAM

```
import pandas as pd

df = pd.read_csv("computer.csv")
df = df[df["Purchase"].notna()]
concepts = df.iloc[:, :-1].values
target = df.iloc[:, -1].values
hypothesis = None

for i in range(len(target)):
    if target[i] == "Yes":
        if hypothesis is None:
            hypothesis = concepts[i].copy()
    else:
        for j in range(len(hypothesis)):
            if hypothesis[j] != concepts[i][j]:
                hypothesis[j] = "?"

print("Most Specific Hypothesis:")
print(hypothesis)

OUTPUT

Most Specific Hypothesis:
['?' '?' '?' 'Dedicated']
```

